Raphael Karamagi

919-423-8015 | raphael.karamagi@duke.edu | linkedin.com/in/raphaelkar | github.com/raphaelkaramagi

Education

Duke University — Aug. 2025 – May 2029
B.S.E. Electrical & Computer Engineering, B.S. Computer Science, GPA: 3.83/4.00 — *Durham, NC*
Selected Coursework: Computer Architecture, Data Structures & Algorithms, Signals & Systems, Probability, Linear Algebra, Multivariable Calculus

Experience

Duke University OIT — Aug 2026 – Present
Student Developer – Innovation Co-Lab — *Durham, NC*
- Architecting an LLM-agent knowledge system for the Co-Lab — schema design, hybrid BM25/vector retrieval, MCP tooling, cross-platform desktop client — based on Karpathy's LLM wiki design pattern.
- Hold office hours and run technical workshops, troubleshooting and infrastructure issues for students and faculty.

Duke University — May 2026 – July 2026
Software Engineer Intern – Code+ Program — *Durham, NC*
- Built an ML model security & safety scanning pipeline—spanning model weights, dependencies, and inference red-teaming—producing risk-scored reports before 50+ ML models are deployed to 20,000+ Duke faculty and students.
- Deployed as a containerized platform backed by Postgres, GitLab CI pipelines, and OIDC user authentication.

IntelliSOFT Consulting — July 2024 – Aug. 2024
Software Engineer Intern — *Nairobi, Kenya*
- Implemented REST API endpoints with role-based access in OpenMRS EMR; led deployment to 20+ hospitals across South Sudan and 3 Nairobi pilot sites with 98% uptime.

Projects

Stock Prediction Platform v2 | *Python, PyTorch, JavaScript, C++, FinBERT* — Jan. 2026 – Present
- Led a 5-person team building a dual-model (LSTM + NLP) stock direction prediction system across 20 tickers; achieved 55% and 64% test accuracy on next day movement & volatility respectively.
- Currently architecting v2 around an event-driven backtester with order book simulation, transaction cost modeling, and walk-forward validation

Compost Now Wash Automation | *Arduino, C/C++, MOSFET* — Jan. 2026 – Apr. 2026
- Designed embedded control circuit for indsutrial bin washers (Arduino + 12V solenoid via MOSFET switches, LED beacons, user inputs), saving 200gal of water/day; wrote non-blocking C++ firmware.

ASL Computer Vision Classifier | *PyTorch, OpenCV* — Oct. 2025 – Dec. 2025
- Built a dual-model CV system classifying 29 ASL gestures from 87K+ images (71% test accuracy); designed Landmark NN (∼10K parameters) for real-time, local CPU inference with 100% hand-detection accuracy.

Technical Skills

Languages: Python, Java, C/C++, R, SQL, JavaScript
ML & Data: PyTorch, TensorFlow, scikit-learn, OpenCV, pandas, NumPy
Frameworks & Tools: Flask, React, Node.js, PostgreSQL, Git, Docker, Linux, OAuth

Leadership & Activities

Duke Applied Machine Learning – Senior Engineer — Sep. 2025 – Present
- Mentor junior members through the AITP program; lead Stock Prediction Platform team.

National Society of Black Engineers (NSBE) – Duke Chapter — Aug. 2025 – Present
- Represented Duke at Fall 2025 Regional Conference and 2026 Annual Convention (Baltimore, MD).

Honors & Awards

5th Globally, Cambridge AS-Level Economics | UKMT Senior Math Challenge Silver (2024) | SAT 1540/1600